Interaction of two charged colloidal spheres

Complex disordered systems: Colloids

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Consider a colloidal particle of total charge Q located at the origin, immersed in a solvent of permittivity $\varepsilon = \varepsilon_0 \varepsilon_r$. The solvent contains a symmetric electrolyte with two mobile ionic species:

- Cations of charge $+ze$ and bulk concentration n_0,
- Anions of charge $-ze$ and bulk concentration n_0.

1. Show that the mean electrostatic potential $\phi(\mathbf{r})$ satisfies Poisson's equation:

$$\nabla^2 \phi(\mathbf{r}) = -\frac{1}{\varepsilon} \left[ze\, n_+(\mathbf{r}) - ze\, n_-(\mathbf{r}) + Q\delta(\mathbf{r}) \right].$$

   Assuming the ions form an ideal gas in thermal equilibrium, their local densities follow Boltzmann distributions:

$$n_+(\mathbf{r}) = n_0 \exp[-\beta ze\, \phi(\mathbf{r})], \qquad n_-(\mathbf{r}) = n_0 \exp[+\beta ze\, \phi(\mathbf{r})],$$

   where $\beta = 1/(k_B T)$.

2. Use these expressions to obtain the (simplified) *nonlinear Poisson–Boltzmann equation* :

$$\nabla^2 \phi(\mathbf{r}) = \frac{2zen_0}{\varepsilon} \sinh\left(\beta ze\, \phi(\mathbf{r})\right) - \frac{Q}{\varepsilon}\delta(\mathbf{r}).$$

3. Find physical conditions to linearise the equation.

4. Solve the resulting linear equation for spherical symmetry to obtain the screened Coulomb (Debye–Hückel) potential:

$$\phi(r) = \frac{Q}{4\pi\varepsilon} \frac{e^{-\kappa r}}{r}, \qquad \kappa^2 = \frac{2n_0 z^2 e^2}{\varepsilon k_B T}.$$

The parameter κ^{-1} is the Debye screening length, and the solution shows that ionic screening converts the long-range Coulomb potential into a short-range Yukawa potential.

The following real to Fourier space mappings may be useful

Real space	Fourier space
$f(\mathbf{r})$	$f(\mathbf{k})$
$\nabla^2 f(\mathbf{r})$	$-k^2 f(\mathbf{k})$
$\delta(\mathbf{r})$	1
$\dfrac{e^{-\kappa r}}{4\pi r}$	$\dfrac{1}{k^2+\kappa^2}$

Solution

We start by considering a single colloidal particle of total charge Q located at the origin, immersed in a solvent characterized by a permittivity $\varepsilon = \varepsilon_0 \varepsilon_r$. The solvent contains a symmetric electrolyte composed of two mobile ionic species: cations of charge $+ze$ and bulk concentration n_0, and anions of charge $-ze$ with the same bulk concentration n_0. The overall system is globally electroneutral in the bulk, so the net ionic charge far from the colloid is zero. The presence of the charged colloid generates an electrostatic potential $\phi(\mathbf{r})$ in the surrounding medium.
The mean electrostatic potential satisfies Poisson's equation, which relates the divergence of the electric field to the local charge density. In this system, the total charge density consists of the fixed colloidal charge, represented by $Q\delta(\mathbf{r})$, and the contribution from the mobile ions, which depends on their local concentrations. This gives

$$\nabla^2 \phi(\mathbf{r}) = -\frac{1}{\varepsilon} \left[ze\, n_+(\mathbf{r}) - ze\, n_-(\mathbf{r}) + Q\delta(\mathbf{r}) \right].$$

To relate the ionic densities to the electrostatic potential, we assume the ions behave as an ideal gas in thermal equilibrium. Each ion experiences a potential energy proportional to its charge times the local potential, $U_\pm(\mathbf{r}) = \pm ze\phi(\mathbf{r})$. In equilibrium, the number density of each species follows the Boltzmann distribution, so the cation and anion densities are

$$n_+(\mathbf{r}) = n_0 \exp[-\beta ze\, \phi(\mathbf{r})], \qquad n_-(\mathbf{r}) = n_0 \exp[+\beta ze\, \phi(\mathbf{r})],$$

where $\beta = 1/(k_B T)$ is the inverse thermal energy. This captures the intuitive fact that cations are repelled by positive potentials and attracted to negative potentials, while anions behave oppositely.
Substituting these expressions for n_+ and n_- into Poisson's equation yields

$$\nabla^2 \phi(\mathbf{r}) = -\frac{1}{\varepsilon} \Big[zen_0 e^{-\beta ze\phi} - zen_0 e^{\beta ze\phi} + Q\delta(\mathbf{r}) \Big].$$

The difference of exponentials can be written as twice a hyperbolic sine, leading to the nonlinear Poisson–Boltzmann equation

$$\nabla^2 \phi(\mathbf{r}) = \frac{2zen_0}{\varepsilon} \sinh\left(\beta ze\, \phi(\mathbf{r})\right) - \frac{Q}{\varepsilon}\delta(\mathbf{r}).$$

This equation describes how the potential generated by the colloid is screened by the surrounding ions. The nonlinearity arises because the ionic response depends exponentially on the potential itself.
The equation can be linearized under the condition that the potential is small compared to the thermal voltage, specifically when $|\beta ze\phi(\mathbf{r})| \ll 1$. In this limit, the hyperbolic sine can be approximated by its argument

$$\sinh(\beta ze\phi) \approx \beta ze\phi.$$

The Poisson–Boltzmann equation then reduces to a **linear** differential equation

$$\nabla^2 \phi(\mathbf{r}) - \kappa^2 \phi(\mathbf{r}) = -\frac{Q}{\varepsilon}\delta(\mathbf{r}),$$

where the inverse Debye length κ is defined by

$$\kappa^2 = \frac{2n_0 z^2 e^2}{\varepsilon k_B T}.$$

This linearized equation is known as the Debye–Hückel equation and describes the potential around a point charge in a screening medium.
This can be solved in Fourier space using the relations:

Real space	Fourier space
$f(\mathbf{r})$	$f(\mathbf{k})$
$\nabla^2 f(\mathbf{r})$	$-k^2 f(\mathbf{k})$
$\delta(\mathbf{r})$	1
$\dfrac{e^{-\kappa r}}{4\pi r}$	$\dfrac{1}{k^2+\kappa^2}$

This gives

$$\phi(r) = \frac{Q}{4\pi\varepsilon}\frac{e^{-\kappa r}}{r}.$$

At short distances ($r \ll \kappa^{-1}$), the potential reduces to the familiar Coulomb form, while at larger distances it decays exponentially due to ionic screening. The characteristic decay length κ^{-1} is the Debye screening length, which increases with decreasing ionic strength and with increasing temperature (at fixed permittivity). This solution illustrates how the electrolyte effectively screens the electrostatic interactions between colloidal particles, converting the long-range Coulomb potential into a short-range interaction that governs the stability and dynamics of colloidal suspensions.

Note. In the linearised regime, the Poisson-Boltzmann equation reads

$$\nabla^2\phi - \kappa^2\phi = -\frac{Q}{\varepsilon}\delta(\mathbf{r})$$

A similar equation exists in field theory for a massive scalar field $\Phi(\mathbf{r})$: it is the Klein-Gordon equation in Euclidean space

$$\left(-\nabla^2 + m^2\right)\Phi(\mathbf{r}) = J(\mathbf{r})$$

It is easy to identify the field the mass and the source with the respective static potential field, the Debye screening parameter and the point charge. The Yukawa potential is then simply the **propagator** (the Green's function).